



భారతీయ సాంకేతిక విజ్ఞాన పరిషత్ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

A Machine Learning Approach For Classifying Black Holes And Neutron Stars Based On Long Term Maxi And Swift Study Of LMXBs

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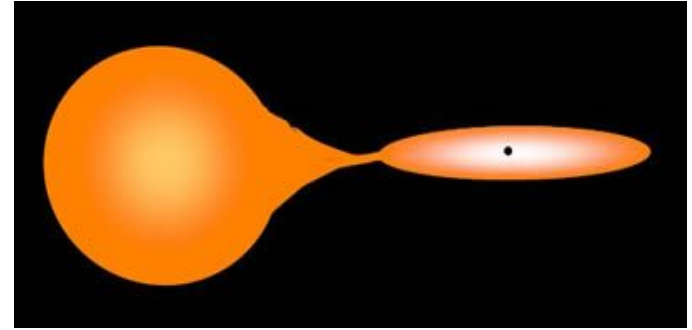
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What Are LMXBs

- Low-Mass X Ray Binaries are binary systems comprising of a compact object (a black hole or neutron star) that accretes gas from a companion star of mass similar to or less than Sun.
- Among the brightest X-ray point sources in the sky when they are actively accreting.
- That is why they are typically discovered and identified during active accretion.
- For many LMXBs the nature of the compact primary remains to be identified.
- Black Holes have been found more frequently in Low-Mass X Ray Binaries, as opposed to high-mass X-ray binaries.
- X-ray binaries with massive companion stars (typically greater than 10 solar mass) are called high-mass X-ray binaries (HMXBs).



Problem Statement

Existing methods to distinguish BH and NS for LMXBs:

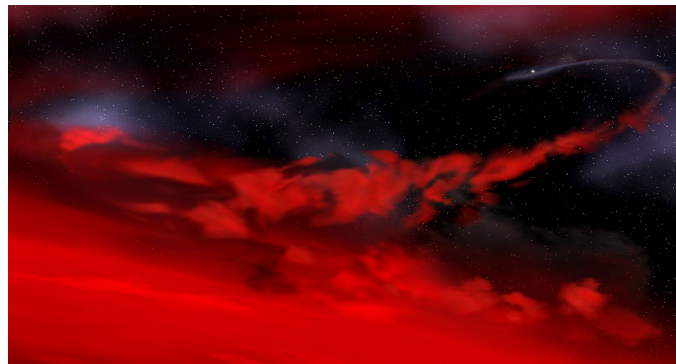
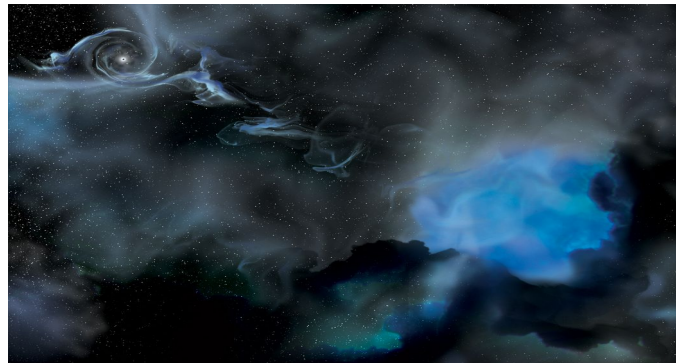
- Spectral fitting (power-law tails)
- Luminosity comparison
- Timing features (QPOs)
- Temperature–luminosity relations

Limitations

- Require detailed spectral modelling
- Depend on transient observational signatures
- Not always conclusive
- Strong spectral degeneracies exist

Motivation of this work

Use long-term X-ray light curves and Hardness-Intensity Diagrams (HIDs) to extract statistical features (like spectral state occupation and outburst behaviour) for **model-independent BH/NS classification**



Hardness Ratio and Spectral States

Spectral states in LMXBs

- Hard State
- Soft State

Hardness Ratio (HR) = $F_{\text{hard}}/F_{\text{soft}}$,

where:

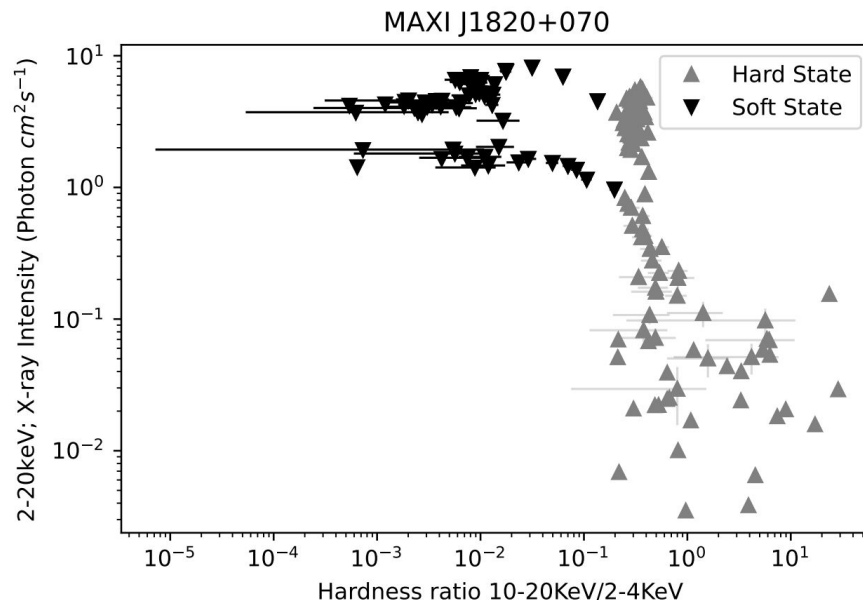
- F_{hard} = hard X-ray flux
- F_{soft} = soft X-ray flux

Physical Interpretation

- Higher HR $\rightarrow$ harder spectrum
- Lower HR $\rightarrow$ softer spectrum

Hardness–Intensity Diagram (HID) tracks:

- spectral evolution
- state transitions
- outburst behaviour



Dataset and Data Sources

Data source

Long-term X-ray monitoring (13 years) from:

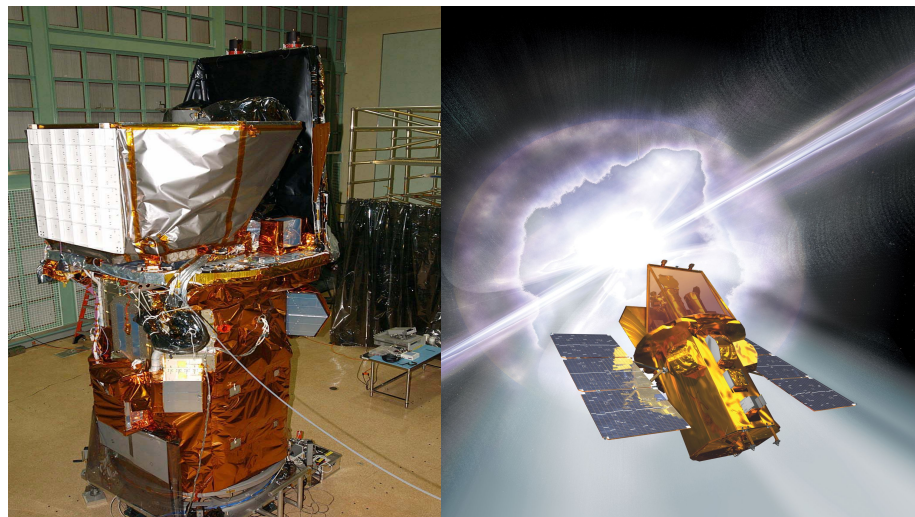
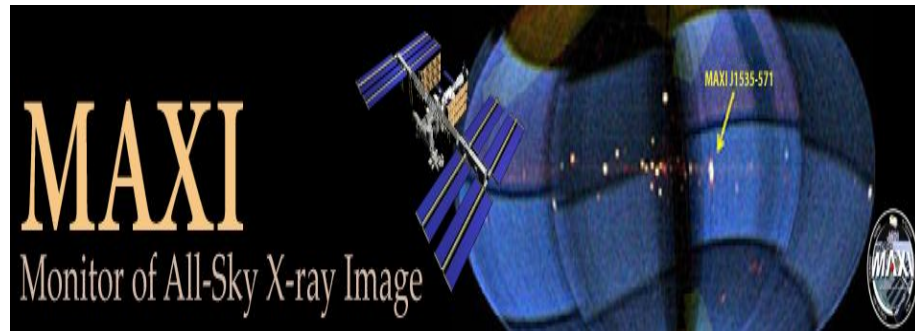
- MAXI
- Swift/BAT

Dataset composition:

- Training dataset:
 - 31 known LMXBs (11 BH and 20 NS)
- Candidate dataset:
 - 11 candidate systems

Data structure

- For each source we have:
 - Time (MJD)
 - Soft flux
 - Hard flux
 - Total intensity
 - Measurement errors



Feature Extraction

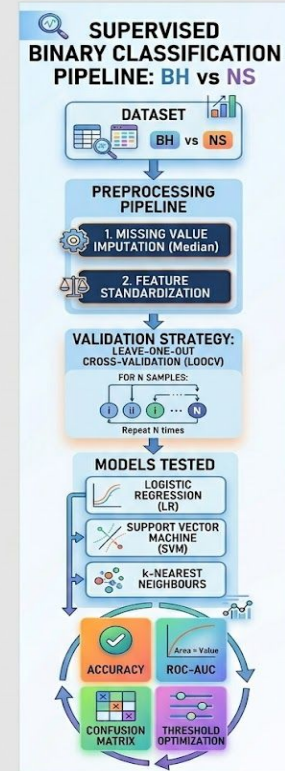
We Convert long-term light curves into physically meaningful features:

- Spectral Features
 - Mean Hardness Ratio
 - Hard state fraction
 - Soft state fraction
- Outburst features
 - Number of outbursts
 - Mean outburst duration
 - Median outburst duration
 - Maximum outburst duration
- Duration distribution features (Fraction of outbursts):
 - 15–60 days
 - 60–100 days
 - 100–200 days
 - > 200 days

We have a total of 11 features for each source

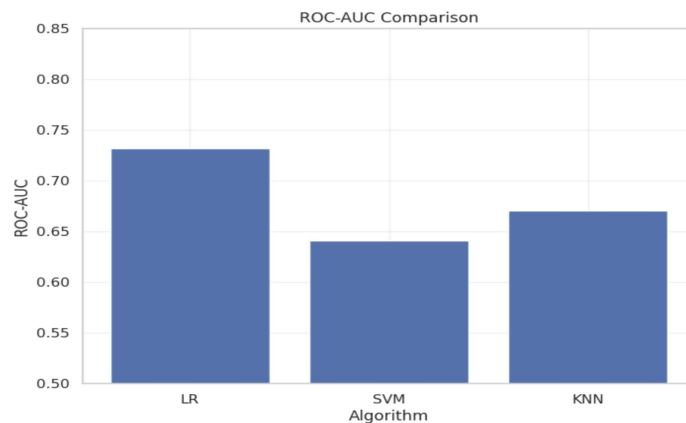
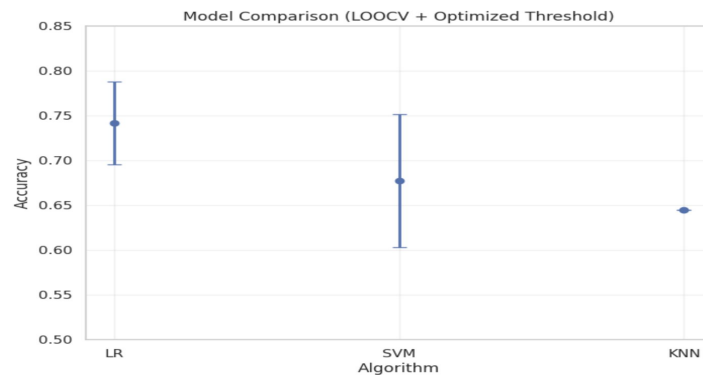
Machine Learning Methodology

- Supervised binary classification : BH vs NS
- Models tested:
 - Logistic Regression (LR)
 - Support Vector Machine (SVM)
 - k-Nearest Neighbours (KNN)
- Preprocessing pipeline
 - Missing value imputation (median)
 - Feature standardization
- Validation strategy
 - Leave-One-Out Cross-Validation (LOOCV)
 - For N samples: Train on N-1 and Test on 1
 - Repeat N times
- Performance metrics
 - Accuracy
 - ROC-AUC
 - Confusion Matrix
- Threshold optimization



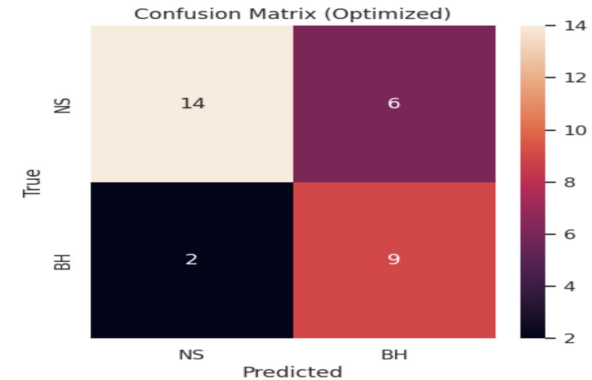
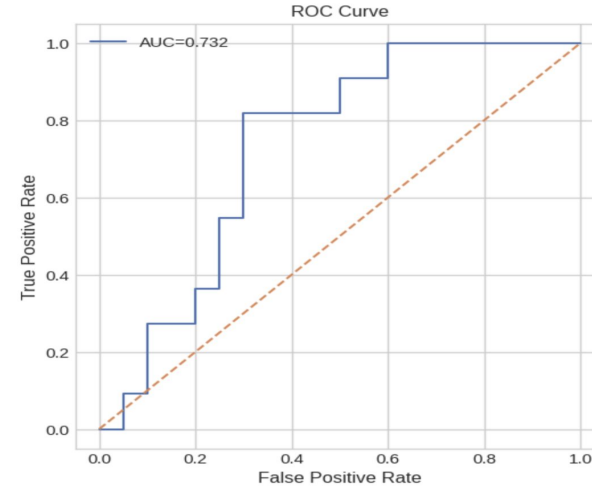
Model Comparison

- Models tested
 - Logistic Regression (LR)
 - Support Vector Machine (SVM)
 - k-Nearest Neighbours (KNN)
- Key observations
 - LR achieves highest accuracy (~74%)
 - LR achieves highest ROC-AUC (~0.73)
 - Better probability stability than SVM and KNN
- Final choice
 - Weighted Logistic Regression selected as the final model



Logistic Regression Performance

- Performance metrics
 - Accuracy: 74.19%
 - ROC-AUC: 0.73
- Confusion Matrix
 - Correctly classified:
 - NS = 14
 - BH = 9
 - Misclassified:
 - NS → BH = 6
 - BH → NS = 2
- Key observation
 - Model shows meaningful separation between BH and NS systems.



Threshold Optimization

Default classification threshold:

- $P(BH)=0.5$ may not be optimal.

Optimized threshold:

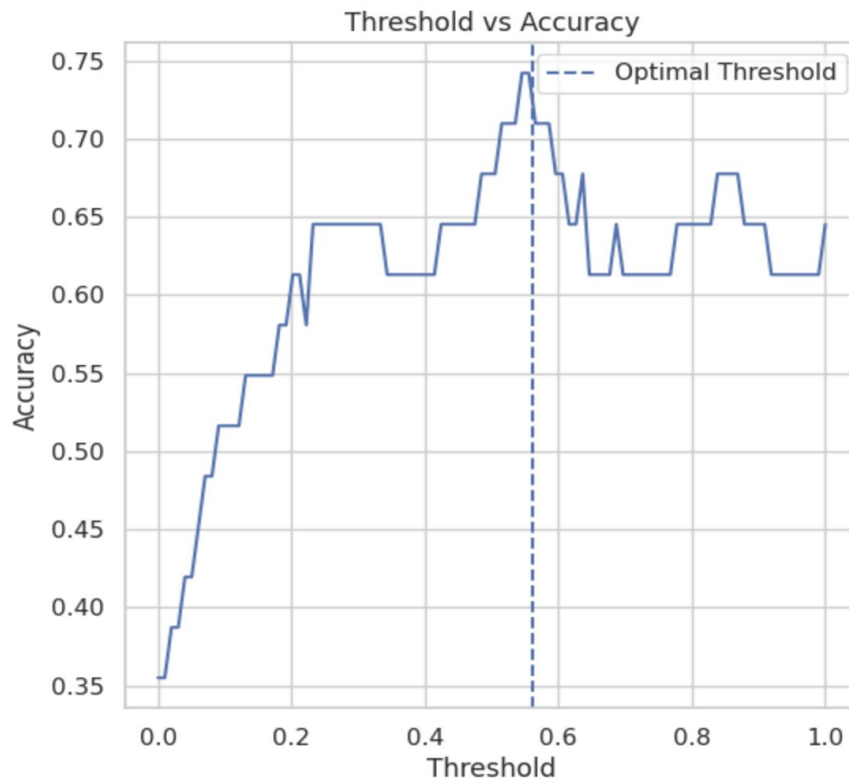
- Best threshold: $P(BH)=0.56$

Effect :

- improves classification accuracy
- better adapts to class overlap
- provides better decision boundary

Key result:

- Probability-based classification is more effective than strict binary thresholding.



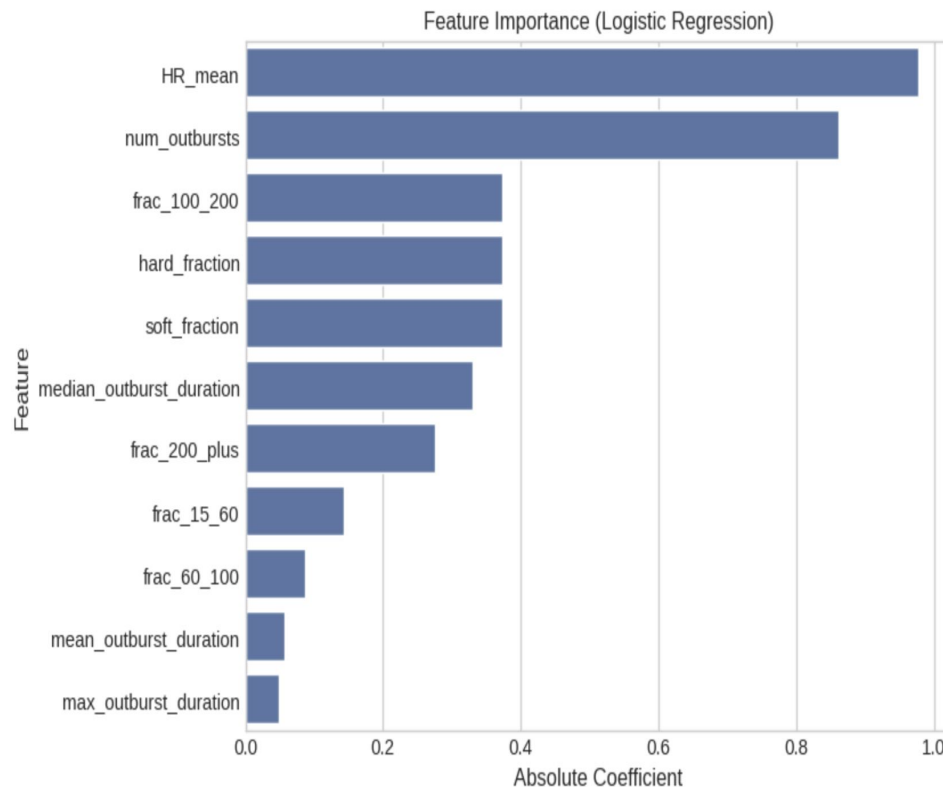
Feature Importance

Top contributing features:

- Mean Hardness Ratio
- Number of outbursts
- Fraction of outburst in 100-200 days
- Hard state fraction
- Soft state fraction
- Median Outburst Duration
- Fraction of outburst greater than 200 days

Observation:

- Hardness-based features dominate classification.
- Outburst behaviour provides additional information.



Physical Interpretation of Important Features

- **Mean Hardness Ratio**
 - Measures average spectral hardness
 - NS systems generally show harder spectra
 - Boundary layer emission adds hard X-ray contribution
 - Strong BH/NS discriminator
- **Number of Outbursts**
 - Captures long-term activity frequency
 - NS systems often show repeated shorter activity episodes
 - BH systems may show fewer but stronger outbursts
 - Encodes variability behaviour
- **Fraction of Outbursts (100–200 days)**
 - Represents intermediate-long outburst population
 - BH systems often sustain longer accretion episodes
- **Hard State Fraction**
 - Fraction of time spent in hard spectral state
 - NS systems spend significant time in hard states
 - Reflects accretion geometry

Physical Interpretation of Important Features

- **Soft State Fraction**
 - Fraction of time spent in soft state
 - BH systems frequently enter strong soft states
 - Indicates disk-dominated emission
- **Median Outburst Duration**
 - Characterizes typical outburst timescale
 - More robust than mean duration
 - Captures source-specific temporal behaviour
- **Fraction of Outbursts (>200 days)**
 - Tracks very long outburst events
 - Indicates prolonged mass transfer and stable accretion
 - Strong signature of long-duration BH transients

Final interpretation:

- Model is learning:
 - spectral hardness evolution
 - state occupancy
 - long-term accretion behaviour
 - not arbitrary statistical patterns.

Candidate System Classification

Confidence-based classification

We define four probability regions:

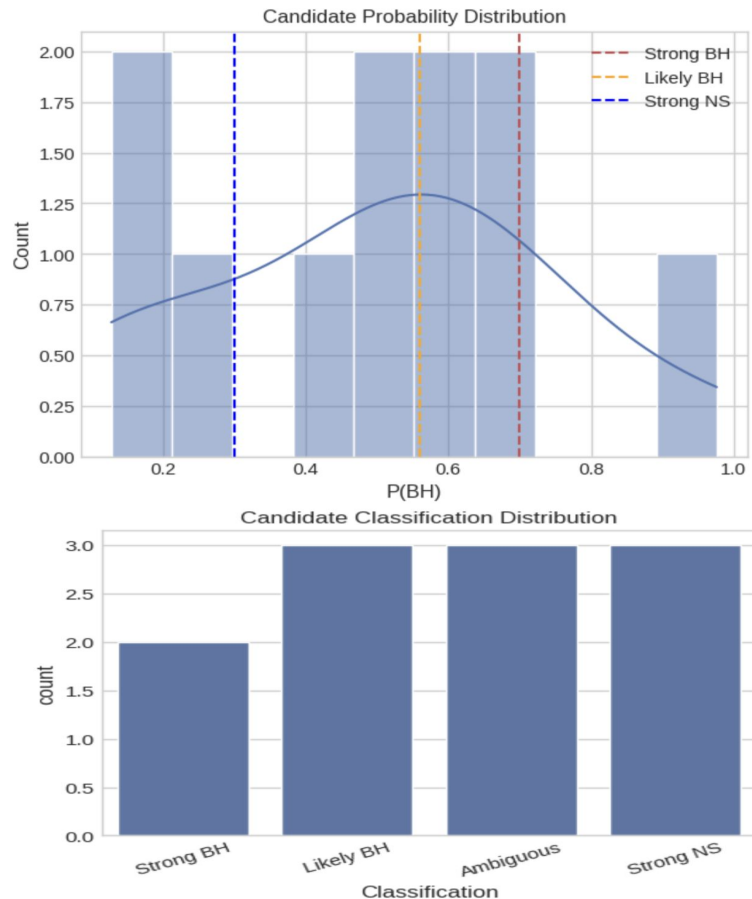
- **Strong BH** $\rightarrow P(\text{BH}) \geq 0.7$
- **Likely BH** $\rightarrow 0.56 \leq P(\text{BH}) < 0.70$
- **Ambiguous** $\rightarrow 0.3 < P(\text{BH}) < 0.56$
- **Strong NS** $\rightarrow P(\text{BH}) \leq 0.3$

Key result:

- MAXI J1848-015:
 - $P(\text{BH})=0.127$ classified as **Strong NS**
 - **Consistent** with observational studies

Observations:

- High-confidence BH candidates identified
- Ambiguous systems remain near decision boundary
- Some candidate BHs classified as NS



Learning Curve and Limitations

Observation:

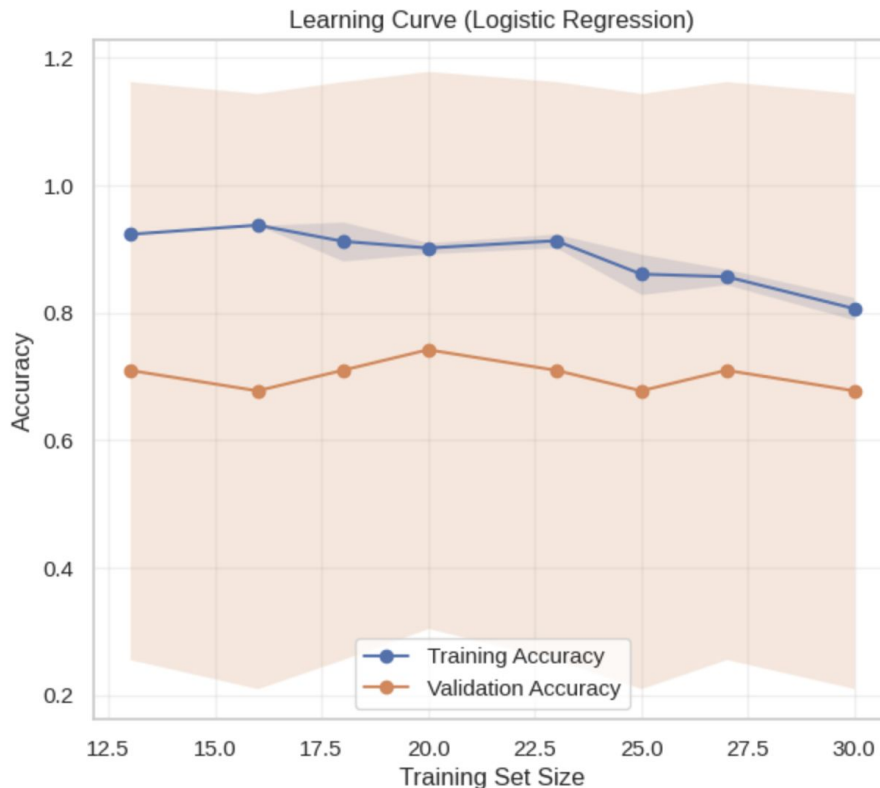
- Training score remains high
- Validation score improves with more data
- Gap between curves still exists

Interpretation:

- Model is **data-limited**, not algorithm-limited.
- More training data may improve:
 - Accuracy
 - Stability
 - Generalization

Current Limitations:

- Small dataset (31 training sources)
- Overlap in BH and NS feature space
- No detailed spectral modelling



Conclusion

- Developed a machine learning framework for BH/NS classification.
- Used physically interpretable spectral and temporal features.
- Logistic Regression performed best among tested models.
- Achieved:
 - Accuracy=74%, ROC-AUC=0.73
- Key result:
 - Correctly identified- MAXI J1848-015 → Strong NS (validates model physically)
- Main insight:
 - Classification is inherently **probabilistic**, not deterministic.
 - BH and NS systems show significant observational overlap.
- Machine learning can provide a useful complementary tool for compact object classification in X-ray binaries.



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**Thank
You!**